Convenience Store Product Recommendation Model

Business Question

Which products should a convenience store stock at different times of day, based on the demographics and behavior of the store's surrounding area? This model groups stores into segments by neighborhood demographics (income, poverty rate, commute patterns) and real sales behavior, then identifies the top-selling products for that segment at a given time of day, giving store owners and executives a concrete, data-driven answer to what to stock and when.

Feature Table (X)

The model uses demographics pulled from U.S. Census ZCTA-level data (matched to each store's ZIP code) combined with real store sales behavior. Demographics were used for both clustering stores into segments and predicting a store's segment; behavioral features were added to the prediction step only, since they meaningfully improve accuracy for stores with existing sales history.

Feature	Source
Median household income ($)	ZCTA-level Census (ACS5)
Per-capita income ($)	ZCTA-level Census (ACS5)
Poverty rate (%)	ZCTA-level Census (ACS5 Profile)
Average commute time (min)	Computed from Census commute-time buckets
Population	ZCTA-level Census (ACS5)
Total revenue ($)	Store transaction history
Average basket size ($)	Store transaction history
Total transactions	Store transaction history
Distinct categories sold	Store transaction history
Total discount amount ($)	Store transaction history
Fuel-to-instore conversion rate	Store transaction history

Target Variable (Y)

The model's target is the store's segment: one of 7 groups produced by K-means clustering on demographics. Because income and poverty rate were highly right-skewed, a handful of very wealthy or very high-poverty ZIP codes distorted a single combined clustering, stores were first split into “typical” and “extreme” groups using the standard 1.5×IQR outlier rule, then clustered separately: 5 segments among typical stores (214 stores) and 2 segments among extreme stores (25 stores), combined into one final 7-class label (e.g. typical_0, typical_1, ..., extreme_0, extreme_1). Each segment's real transaction history was then used to identify its top-selling branded products by time of day (morning, afternoon, evening, night) - fuel, cigarettes, tobacco, alcohol, and lottery were excluded from recommendations, since these are not general merchandising decisions.

Model & Methodology

1. Cluster : K-means grouped stores by demographics only (income, poverty rate, commute time, population), separately for typical and extreme stores, then combined into 7 segments.

2. Predict : A Decision Tree Classifier was trained to predict a store's segment from its demographics and behavioral data, so the model can also classify a new location before it has any sales history.

3. Recommend : Each segment's actual transaction data (branded item level, joined to the product master table) was ranked by units sold, by time of day, to produce the final recommendation list.

Model performance: 83.3% test accuracy, against a 21.8% majority-class baseline. The model performed reliably across the 5 typical-store segments (F1-scores 0.82–0.90); the 2 extreme segments showed promising results (F1 0.67–0.80) but were evaluated on very few test examples (2–3 stores each) given their smaller size, so those results should be treated as directional rather than statistically firm.

Chart 1 : Top Recommended Product by Segment and Time of Day

This is the model's core output: for every segment, the single best-selling (non-restricted) branded product at each time of day, based on real transaction data.

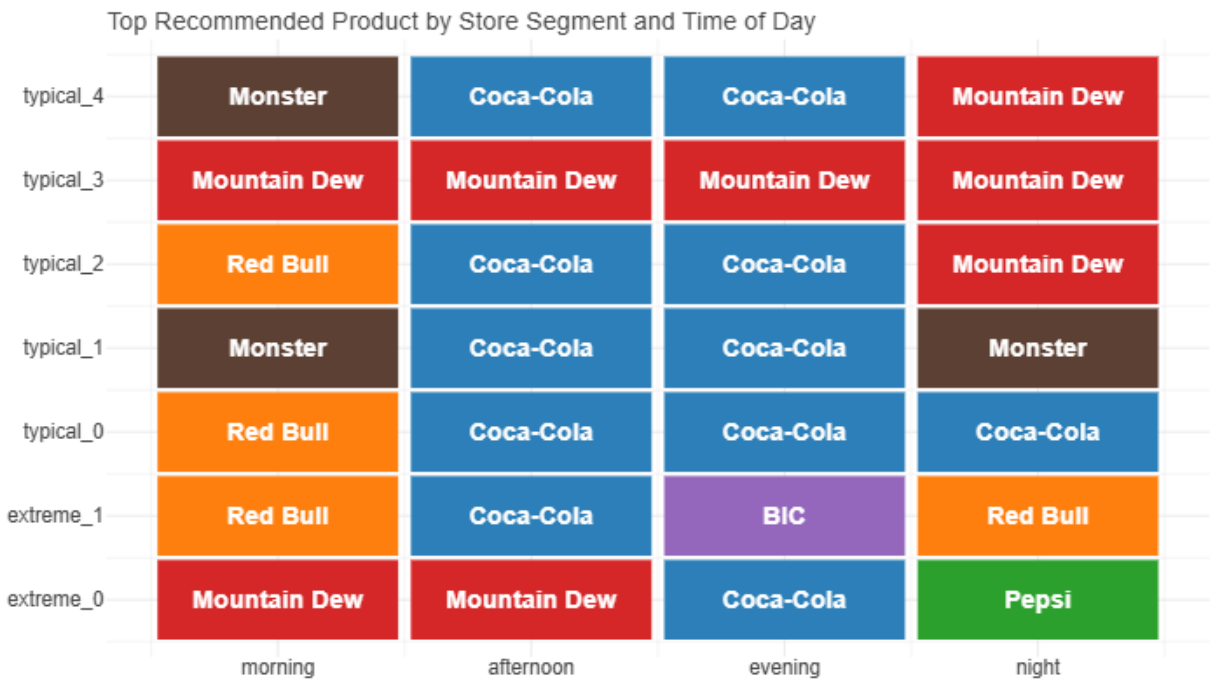
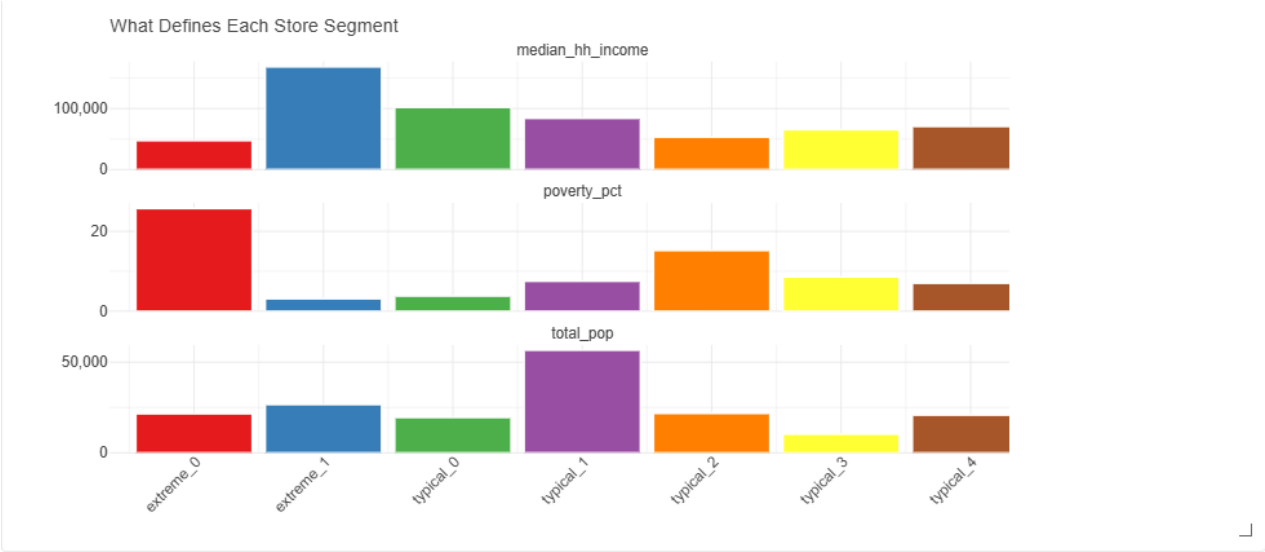


Chart 2 : What Defines Each Store Segment

This chart shows why the segments differ — average commute time, median household income, and poverty rate across the 7 segments. The two extreme segments (leftmost two bars in each panel) are visibly distinct from the five typical segments, confirming the typical/extreme split captured a real structural difference in the data rather than an arbitrary cut.



Business Recommendations

1. **Stock by time of day, not just by store.** The top-selling product changes across morning, afternoon, evening, and night within the same segment. Knowing this ahead of time means a store can order the right products in bundles instead of guessing. Buying in bulk for what actually sells at a given time is cheaper than ordering piece by piece and hoping it moves.

2. **Treat "extreme" segments as genuinely different businesses, not outliers to ignore.** High-poverty stores (extreme_0) and high-income stores (extreme_1) sit far outside the range of the other 5 segments on income and poverty rate. Applying a "typical store" promotion strategy to either is likely to underperform; extreme_0 locations may respond better to value-sized/budget bundles, while extreme_1 locations may support premium or convenience-priced items with less price sensitivity.

3. **Use the segment model when scouting new locations.** Because the model predicts segment from public demographic data (income, poverty rate, commute time, population) plus estimable behavior, a new site can be classified into a segment and given a starter product mix before a single transaction happens, shortening the ramp-up period for a new store.

Business Recommendations

Databricks notebook (https://dbc-eb84d3a1-867f.cloud.databricks.com/editor/notebooks/880022905244856?o=7474648749221721#command/7773840513282225)